

ARSQUARE EDUCATIONAL CONSULT

MOCK ENTRANCE EXAMINATION **(MATHEMATICS)**

1. The sum of two numbers is 156. Their difference is 18. Find the smaller number A.)84 B.)69 C.)120 D.)57 E.)40
2. What is the value of 16.0681 correct to two decimal places A.)16.07 B.)20 C.)13.45 D.)7.45 E.) 10.30
3. The angle between the hand of a clock at 3:00 clock is A.)90 B.)360 C.)180 D.)0 E.)36
4. A motorist travelled a distance of 60km in 1hour 30mins. What is his speed in km per hour A.)91 B.)18 C.)4 D.)45 E.)40
5. In a society of 100 members, 60% of them paid 10,000 each and the rest paid 15,000 into a charity appeal fund. How much did they contribute? A.)300,000 B.)350,000 C.)500,000 D.)1,200,000 E.)450,000
6. What does the roman numeral MCMLXIV represent? A.) 1196 B.)1964 C.)1945 D.)10096 E.)1795
7. A circle has an area which is 4840sq,m. What is its radius? A.)35.5m B.)39.2m C.)42.9m D.)48.4m E.)59.3m
8. The area of a Garden, square in shape is 676sq.m How long is its side? A.)10m B.)26m C.)52m D.)34m E.)71m
9. The area of a circle is 154cm square, what is the circumference of the circle. A.)22cm B.)44cm C.)77cm D.)88cm E.)308cm
10. The volume of a rectangular tank is 420 metre cube, if the length and height of the tank are 14m and 5m respectively, what is the width of the tank? A.)6m B.)16m C.)30m D.)70m E.)84m
11. A book has 25 lines per page. Each line has an average of 8 words. How many are in this book if it contains 350 pages? A.)7000 B.)70,000 C.)700 D.)700,000 E. 7,000,000
12. What is two-third of the sum of 19 and 17 A.)24 B.)72 C.)12 D.)36 E.)18

13. Convert 37_{ten} to binary number A.)10110 B.)100101
C.)111000 D.)100100
14. A rectangle is 3m long and it's perimeter is 11m. Find it's area. A.)2.5 B.)6 C.)7.5 D.)10.5 E.)15
15. Simplify $7x - 4x + 3x + 6x$ A.)7x B.)9x C.)10x D.)12x E.)14x
16. What is the median of the following numbers:9,12,10,9,12,16,9,14? A.)12 B.)-12 C.)15 D.)20 E.)-20
17. The mode of the data in the question above is?A.)9 B.)10
C.)11 D.)12 E.)14
18. Express 2500 centimeter cube in litres A.)2.5 litres B.)25
litres C.)250 litres D.)2500 litres E.)25000 litres
19. Find the simple interest paid on 2,400 for 6 months at 3%
rate per annum A.)7.5 B.)36 C.)42.5 D.)140 E.)240
20. Write the number 2014 in roman numerals A.)MCMXXXIV
B.)MMCXXIV C.)CMMXXXVI D.)MMXIV E.)MMXIII
21. Divide the LCM of 24,28,36,50 by the product of 2 and 3
A.)700 B.)820 C.)1200 D.)2100 E.)12600
22. Which of the following is the lowest? -3,-5,0,-18,-14,-7 A.)-
18 B.)-14 C.)-7 D.)-5 E.)0
23. If 6 pencils cost 10.80 naira , how much would 13 pencils
cost? A.)23.40 B.)23.90 C.)25.90 D.)26.90 E.)27.90
24. A piece of ribbon is cut up in the proportion 2:3:4, if the
greatest piece is 16cm,then the length of the original piece of
ribbon is? A.)24cm B.)36m C.)42cm D.)63cm E.)144cm
25. A wristwatch which cost 300 naira was sold at a profit of
20%, its selling price was? A.)200 B.)200.40 C.)360 D.)204
E.)204.80
26. A cube of 4 added to a cube root of 8 is? A.)60 B.)62 C.)66
D.)80 E.)81
27. A polygon with eight sides is called? A.)nonagon
B.)decagon C.)octagon D.)pentagon E.)henagon
28. Which of the following numbers does not belong to the
group? A.)1 B.)11 C.)21 D.)31 E.)51
29. The roman numeral CXCV represents? A.)194 B.)196
C.)214 D.)215 E.)216
30. The following are quadrilaterals except? A.)circle
B.)parallelogram C.)rectangle D.)square E.)trapezium
31. Which of the following shapes has four lines of symmetry?
A.)square B.)cube C.) rectangle D.)kite E.)parallelogram
32. Write ten thousand, one hundred and ninety in figures.
A.)1090 B.)10191 C.)10119 D.)10019 E.)10190

33. Find the product of the prime numbers between 1 and 6.
A.)10 B.)15 C.)24 D.)30 E.)48
34. A car travels 70km on 8 litres of petrol , how far will it travel on 12 litres? A.)117km B.)108km C.)85km D.)50km
35. Find the smallest number that will give remainder 2 when divided by 4,6 and 9 A.)58 B.)54 C.)38 D.)26
36. Find the least number which is divisible by 6,8 and 9 without any remainder A.)72 B.)54 C.)48 D.)36
37. What are the prime factors of 60. A.)1,2,7 B.)2,5,7 C.)1,3,5 D.)2,3,5
38. The sum of two numbers is 50. The second number is four times the first. Find the smaller number A.)25 B.)20 C.)15 D.)10
39. Simplify $3(5y-2x)-7(4y-x)$ A.) $13x+43y$ B.) $-13x+13y$ C.) $x-13y$ D.) $5x-13y$
40. Round off 0.009238 to 3 significant figures A.)0.01 B.)0.009 C.)0.00923 D.)0.00924
41. How many perfect squares are in the set (1,3,4,6,9,64,67)
A.)1 B.)2 C.)3 D.)4
42. What is the place value of 5 in the number 249.056? A.)five tenths B.)five hundredths C.)five D.)fifty
43. What is 0.002568 correct to 3 decimal places? A.)0.00 B.)0.002 C.)0.00257 D.)0.003
44. Decrease 2000 by 20%. A.)1,600 B.)2,400 C.)1,980 D.)1,800
45. Change 0.7 to percentage A.)0.07% B.)0.7% C.)7% D.)70%
46. If 45% of a number is 126, find the number A.)280 B.)235 C.)145 D.)126
47. How many seconds are in 123 minutes? A.)60 seconds B.)100 seconds C.)105 seconds D.)154 seconds
48. Find the capacity of a tank 50cm by 40cm by 30cm in litres
A.)60 litres B.)62 litres C.)70 litres D.)73 litres
49. Evaluate 89^2-11^2 A.)8900 B.)9000 C.)9980 D.)7800
50. A rectangular field is 7 metres in width, its length is 5 metres longer than its width. What is the perimeter A.)12m B.)19m C.)38m D.)84m
51. The ratio of the radii of two wheels is 4:5. Find the ratio of their circumference A.)4:5 B.)4:25 C.)8:25 D.)16:25
52. Find the area of a circle whose circumference is the same as the perimeter of a square of side 22cm. A.) 616cm^2 B.) 308cm^2 C.) 154cm^2 D.) 77cm^2
53. An odometer is an instrument used to measure what in automobiles? A.)speed B.)odour C.)direction D.)distance

54. How long will it take a train 55m long to cross a bridge 55m long if it travels 55m in 5 minutes A.)20 minutes B.)11 minutes C.)10 minutes D.) 5 minutes
55. Mrs Olowe left lagos railway terminus at 4.30pm to attend a crucial meeting. The journey took her 12 hours. At what time did she arrive at sokoto the next day? A.)2.30am B.)3.30am C.)4.30am D.)5.30am
56. Decrease 448 by 25% A.)102 B.)336 C.)112 D.)560
57. Given that $x=2$ and $y=4$, find the value of $3x+y$ A.)10 B.)14 C.)20 D.)24
58. If 12 men can cut a tree in 4 days, how many days can 4 men cut the same tree if they work at the same rate. A.)8 B.)12 C.)10 D.)4
59. Find the LCM of 40,48 and 24 A.)140 B.)120 C.)160 D.)240
60. If $a*b=a+b-ab$, what is the value of $8*5$ A.)13 B.)-18 C.)40 D.)-27

GENERAL PAPER MOCK ENTRANCE EXAMINATION

- Which of the following is a core value that promotes unity in the family? A.)Selfishness B.)Tolerance C.)Dishonesty D.)Conflict
- Nigeria is located in which part of Africa? A.)North Africa B.)East Africa C.)West Africa D.) South Africa
- The first capital of Nigeria before it was moved to Abuja was. A.)Ibadan B.)Calabar C.)Lagos D.)Lokoja
- Which arm of the government is responsible for making laws A.) The Executive B.)The Judiciary C.)The Legislature D.)The Police
- The Nigeria National flag was designed by A.)Herbert Macaulay B.)Taiwo Akinkunmi C.)Funmilayo Ransome-Kuti D.)Nnamdi Azikwe
- The planet closest to the Sun is A.)Venus B.)Earth C.)Mercury D.)Mars
- What is the boiling point of pure water A.)0°C B.)50°C C.)80°C D.) 100°C
- Whch of these animals is a mammal? A.)Tilapia B.)Whale C.)Crocodile D.)Eagle

9. Which of the following is the largest desert in the world?

A.)Kalahari B.)Sahara C.)Gobi D.)Namib

10. A push or pull on an object is known as. A.)Energy

B.)Friction C.)Force D.)Gravity

ENGLISH LANGUAGE MOCK

ENTRANCE EXAMINATION

COMPREHENSION PASSAGE

The next man to appear before the judge was thin old man.

“My lord, I beg of you, please reduce my tax assessment,” he said as he prostrated himself metres away from the place in the court assigned to complainants.

“I am an old man. My children have all deserted me to go and trade in the Northern part of the country. I don’t hear from them at all. They don’t send me money. Where can I find Two hundred thousand naira that I am told is written on this paper?” and he brought out from his pockets a creased piece of paper.

The judge whispered while the court house babbled at the case of the old man whose sons has deserted. The presiding judge declared that the tax assessment should be reduced from two hundred thousand naira to Fifty naira. “Clerk, write down Fifty naira“ the judge said.

“I thank you my lord”, said the old man. “ May your fathers in heaven support you with more wisdom. May you dispense justice for a long time. But you should not help me only half the way. When you want to save a lazy man, do you not save him completely? Do you stop at making him half a happy man? So tell the clerk to write not Fifty naira but nothing at all. I am too old to pay taxes. Long may you live.” The judge adjusts his spectacles and said, “Clerk, the man is too old to pay any tax,way too old; write nothing for him. I rise!”

Answer the following questions by choosing the most appropriate option

1. The reaction in the story took place in _____. (a) the oba's palace (b) a court (c) the market place (d) the tax office
2. The old man came to _____. (a) complain about his children (b) complain about his wife (c) visit his friends who are judges (d) complain about his tax assessments.
3. The _____ was asked to write down fifty naira (a) court clerk (b) secretary (c) tax official (d) messenger
4. The _____ declared that the tax should reduce (a) National President (b) Oba of the town (c) court president (d) High court judge
5. The old man said they should _____ (a) help him half the way (b) not help him fully (c) help him fully (d) not help him again
6. The old man was eventually allowed to _____ (a) pay nothing at all (b) pay fifty naira (c) pay two hundred naira (d) settle his disputes with the tax clerk

In each of the following, choose from the options the one which is nearly opposite in meaning to the word underlined

7. His love for movies is second to none. (a) adoration (b) worship (c) dislike (d) disarray
8. A gentleman like him would not intentionally hit another person. (a) unavoidably (b) accidentally (c) aimlessly (d) consciously
9. Attending a secondary school is obligatory for all Nigerian children. (a) voluntary (b) mandatory (c) desirable (d) easy
10. My father rarely travels on sundays. (a) surely (b) nearly (c) never (d) often

In each of the following, choose from the options the one which is nearest in meaning to the word or expression underlined.

11. The width of the hall is about 20 metres. (a) breadth (b) length (c) height (d) size
12. The promotion of literacy amongst our people is the government's major concern. (a) popularity (b) publicity (c) elevation (d) encouragement
13. The main purpose of examination is to assess students's performance. (a) primary (b) biggest (c) secondary (d) tertiary
14. There was a football competition among the schools in the area. (a) game (b) match (c) play (d) rivalry

In each of the following, choose the correct answers from the options given

15. After complaining of tooth ache , his father took him to see a/an_____ (a)optician (b)pharmacist (c)dentist (d)nurse

16. The daughter of one's sister or brother is one's_____ (a)sister (b) niece (c)cousin (d) nephew

17. The match_____ before the governor arrived. (a)had ended (b)had finished (c)is ended (d) is finished

18. The principal praised olamide and _____ for a job well done (a) I (b) me (c) mine (d) my

19. The sentence "she will take the children home" in passive voice becomes "The children....." (a)are taken home by her (b)will be taken home by her (c) was take home by her (d)will be taken home by her

In each of the following, choose correctly the spelt words from the options

20. (a) Ambigious (b)Anbigious (c)Anbiguous (d)Ambiguous

21. (a) Necessary (b) Neccessary (c) Necesary (d) Neccesary

22. (a) Monothong (b)Monopthong (c) Monophthong (d)Monophtong

Choose the most appropriate word or group of words for each gap

For a very long time , I had fantasised about being a ___23___ of the International school, University of Ibadan. Last year, my mother tried to get the ___24___ form but she could not because I just clocked nine years and the school has a ___25___ that discourages underage admissions. Now I am a ten and I can't wait to put on the school uniform of my dreams. For nine weeks, I had prepared for the entrance ___26___ which my brother had informed me would be ___27___. The fact that I had passed the examinations with flying colours was not surprising as I had prepared very meticulously.

- | | A | B | C | D |
|------------|------------------|----------------------|---------------------|-----------------------|
| 23. | Student | ward | child | boy |
| 24. | Education | Entrance | Admission | Academics |
| 25. | Method | procedure | policy | thing |
| 26. | Questions | examination | classworks | projects |
| 27. | On laptop | computer type | google-based | computer based |

Choose the word that has the same vowel or consonant sound as the phonetic symbol given

28. /ɪ/ (a) seal (b) shop (c) church (d)zion

29. /i:/ (a) pi (b) kite (c) pile (d)need

30. /g/ (a) jeer (b) gear (c) giant (d) yam